

Econometrics Stata Lab: Do Smaller Classes Help Kids Learn?

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Week 6, Fall 2026

Running example: Krueger (1999), "Experimental Estimates of Education Production Functions," QJE.

Midterm review: every number in a regression table, on real data.

1. **The question:** Tennessee's class-size lottery (Project STAR).
2. **Idea 1:** a regression on a yes/no variable compares two averages.
3. **Idea 2:** the standard error, t , p and the 95% interval, using 1,000 reruns of Tennessee.
4. **Idea 3:** a tiny R^2 can go with a real effect.
5. **More than one test at once:** three class types and the F -test.

Recap: Week 5 in one slide

- Papke asked whether more money raises Michigan pass rates.
- The answer moved a lot as we changed the comparison:
 $19.6 \rightarrow 22.9 \rightarrow 7.6 \rightarrow 45.1 \rightarrow 6.3$.
- Why? Spending was **not** handed out by lottery. Richer schools, later years and poorer students all came with more money, and each one changed the answer.

Today: a study where a **lottery** decided who got the policy. Watch how little the answer moves.

The question

Smaller classes cost a lot: more teachers, more classrooms. **Do they actually help children learn?**

Comparing schools does not settle it: schools with small classes are often richer, so their children would do well anyway.

Project STAR, Tennessee, 1985

In 79 schools, about 11,600 children and their teachers were put **by lottery** into:

- a **small** class (13–17 children),
- a **regular** class (22–25 children), or
- a regular class **with a teacher's aide**.

Your guess: children score about 920 points (reading + math) and differ by about 75. How much higher do small classes score? 0? 5? 20? 50?

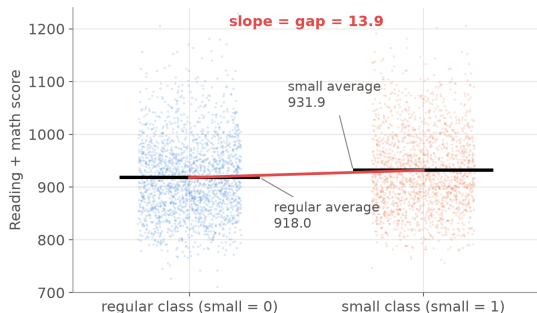
The data: one row per child, one column block per grade

- The file is an **SPSS** file from Harvard Dataverse: `import spss`.
- One row is **one child**: `isid stdntid`.
- The file is **wide**: `gk...` = kindergarten, `g1...` = 1st grade, ..., `g3...` = 3rd grade.

<code>stdntid</code>	<code>gkclasstype</code>	<code>gktreadss</code>	<code>gktmathss</code>	<code>g1classtype ...</code>
12424	2 (regular)	447	520	3
10133	3 (aide)	427	478	3

Today: **kindergarten**, the year of the lottery. `score` = reading + math. `small` = 1 small class, 0 regular class.

Idea 1: a regression on a yes/no variable compares two averages



$$\text{score} = b_0 + b_1 \text{small} + u$$

- $b_0 = 918.0$: average in **regular** classes.
- $b_1 = 13.9$: the **gap**.
- **Fitted value** = the child's group average.
- **Residual** = how far the child is from it.

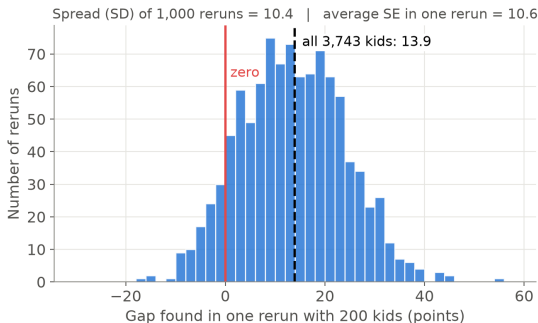
The Stata table

Source	SS	df	MS	Number of obs	=	3,743
				F(1, 3741)	=	32.27
Model	179850.266	1	179850.266	Prob > F	=	0.0000
Residual	20847551.4	3,741	5572.72158	R-squared	=	0.0086
				Adj R-squared	=	0.0083
Total	21027401.7	3,742	5619.29495	Root MSE	=	74.651

score	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
small	13.89899	2.446592	5.68	0.000	9.102211	18.69578
_cons	918.0429	1.667157	550.66	0.000	914.7743	921.3115

Five numbers to read today: **13.9** (the gap), **2.45** (SE), **5.68** (t), **0.000** (p), **[9.1, 18.7]** (95% interval). And one more: $R^2 = \mathbf{0.0086}$.

Idea 2: what if we reran Tennessee?



Pretend we can only test **200** children. Draw 200 real STAR children at random, compute the gap, and repeat 1,000 times.

The reruns **disagree**: from about -18 to $+55$.

Standard error = how spread out the reruns are. One sample's SE (10.6) predicts that spread (10.4).

With all 3,743 children the SE is 2.45: **more data, less wobble**.

t and p in plain words

$$t = \frac{\text{estimate} - 0}{\text{SE}} = \frac{13.9}{2.45} = 5.68$$

- t = how many “wobbles” (SEs) our gap is away from zero.
- **p -value**: *if small classes truly did nothing*, how often would luck alone give a gap at least this far from zero? Here about **1 in 70 million** ($p = 0.00000001$). Stata prints 0.000.
- Rule of thumb: $|t| > 2 \Leftrightarrow p < 0.05 \Leftrightarrow$ “significant at the 5% level”.

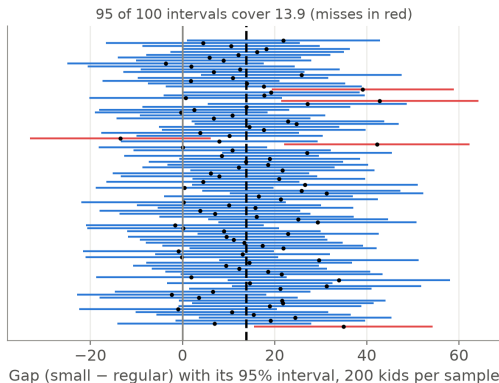
One-sided test: “do small classes *help*?”

$H_0 : b_1 \leq 0$ against $H_1 : b_1 > 0$. Only the upper tail counts, so

$p_{\text{one-sided}} = p_{\text{two-sided}}/2$ when $t > 0$.

Stata: `display ttail(e(df_r), 5.68)`.

What does “95% confidence” promise?



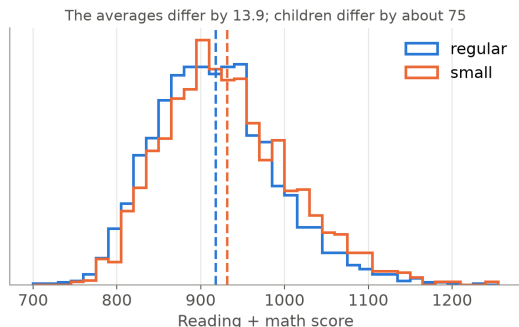
Each bar: one sample of 200 children, $\text{gap} \pm 1.96 \times \text{SE}$.

95 of 100 bars cross the full-data gap (13.9). The red bars missed.

The 95% is a promise about the **method**, not about one bar.

Our real interval, with all children: **[9.1, 18.7]**. Zero is not inside, so we reject “no effect” at the 5% level.

Idea 3: $R^2 = 0.009$, and the effect is still real



$$R^2 = \frac{\text{explained SS}}{\text{total SS}} = \frac{179,850}{21,027,402} = 0.009$$

- Children differ a lot for reasons that have nothing to do with class size.
- Class size explains **less than 1%** of those differences.
- But $t = 5.7$: the gap is real. In SD units it is $13.9/74 \approx 0.19$.

Low $R^2 \neq$ unimportant policy.

The regression ladder: does the answer move?

	(1) Small vs regular	(2) + School dummies	(3) All three types
Small class	13.90*** (2.45)	16.00*** (2.22)	13.90*** (2.41)
Regular + aide			0.31 (2.31)
School dummies	No	Yes	No
Observations	3,743	3,743	5,786
R-squared	0.009	0.238	0.007

(2) School dummies:

13.9 $\rightarrow$ 16.0. Barely moves, because the lottery was held *inside* each school. Compare Papke: no lottery, and the answer jumped around.

R^2 : 0.009 $\rightarrow$ 0.24. Explaining more of the scores is **not** the same as getting the effect right.

(3) **Three class types:** an aide adds 0.3 points ($p = 0.89$).

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Base group: regular class.

More than one test at once: the F -test

regress score ib2.gkclasstype (base group: regular class)

Null hypothesis	Stata	Result
Class type does not matter at all	test 1.gkclasstype 3.gkclasstype	$F = 21.3$, $p = 0.000$
Small class = class with an aide	test 1.gkclasstype = 3.gkclasstype	$F = 32.1$, $p = 0.000$

- One t -test checks **one** coefficient. The F -test checks **several together** (Paul's lecture yesterday).
- The F at the top of every regress table tests "all the x 's are zero".
- Policy reading: shrinking classes helps; adding an aide does not.

What Krueger found

From the paper's abstract (he measures scores in **percentiles**):

- In the first year in a small class, scores rise by about **4 percentile points**.
- The advantage **grows** by about 1 percentile point per year after that.
- Teacher aides and measured teacher characteristics have **little effect**.
- The effect is **larger** for minority children and children on free lunch.

In our scale-score units: $+13.9 \text{ points} = 0.19 \text{ SD}$ in kindergarten.

Take-home in the do-file: reshape long the grades K–3 and check whether the gap lasts (we get 13.9, 19.8, 9.6, 11.7 points); test a free-lunch interaction.

Midterm cheat sheet: reading any regression table

Number	Plain words	STAR
Coefficient	change in y when x goes up by one (for a dummy: the gap)	13.9 points
Std. err.	how much the estimate would move in a rerun	2.45
t	coefficient $\div$ SE: wobbles away from zero	5.68
$P > t $	how often luck alone gives this, if the true effect is 0	0.000
95% CI	coefficient $\pm$ about 2 SE	[9.1, 18.7]
R^2	share of the differences in y that the model explains	0.009
F (top)	test that all x 's are zero together	32.27

Run it in Stata, then play with the notebook

Do-file week06_star.do (change the cd line first):

```
copy "https://dataverse.harvard.edu/api/access/datafile/666716?format=original"
"data/STAR.Students.sav"
import spss using "data/STAR.Students.sav", clear
rename *, lower
keep if flagsgk == 1
generate score = gktreadss + gktmathss
regress score small
regress score ib2.gkclasstype
test 1.gkclasstype 3.gkclasstype
```

Notebook (opens in your browser, nothing to install; link on Canvas):

- rerun Tennessee with any sample size and watch the SE;
- draw 100 confidence intervals and count the misses;
- get a p -value by **shuffling** the class labels.

Reference

Alan B. Krueger (1999), “Experimental Estimates of Education Production Functions,” *Quarterly Journal of Economics* 114(2), 497–532.

In this folder: `Krueger_1999_QJE_Experimental_Estimates.pdf` (published version) and NBER Working Paper 6051 (1997).

Data: C. M. Achilles et al. (2008), “Tennessee’s Student Teacher Achievement Ratio (STAR) Project,” Harvard Dataverse, doi:10.7910/DVN/SIWH9F.

Our sample: kindergarteners in STAR with a reading and a math score (3,743 in small or regular classes; 5,786 including aide classes). $\text{Score} = \text{total reading scale score} + \text{total math scale score (SAT)}$. Krueger uses percentile scores, so his numbers differ slightly.

Tell me how the course is going

- Anonymous, about 1 minute, all questions optional.
- Pick the week, your section, and rate the pace.
- Fill it in once for each week, if you like.

[Open the feedback form](#)



Scan with your phone camera

Appendix: the SE, t , CI and p by hand in Stata

```
regress score small
matrix V = e(V)
display sqrt(V[1,1])
display _b[small] / _se[small]
display invt(e(df_r), 0.975)
display _b[small] - invt(e(df_r),0.975)*_se[small]
display 2*ttail(e(df_r), 5.68)
display ttail(e(df_r), 5.68)
display e(mss) / (e(mss) + e(rss))
ttest score, by(small)
```

variances of the coefficients

$SE = 2.45$

$t = 5.68$

the “about 2” = 1.96

$CI\ low = 9.10$

two-sided p

one-sided p

$R^2 = 0.0086$

same t , opposite sign

Appendix: a p -value by shuffling

1. Take one sample of children and compute the real gap.
2. **Pretend class size does nothing**: shuffle the small/regular labels at random.
3. Compute the gap again. Any gap now is pure luck.
4. Repeat 2,000 times. p = share of shuffles with a gap at least as far from zero as the real one.

In the notebook, with 200 children, the shuffle p and Stata's formula p agree to about two decimals (e.g. 0.027 vs 0.022). With all 3,743 children, no shuffle ever gets close to 13.9.